

Chapter 4

Normal and Injury Mechanics of the Lumbar Spine

Chapter 3 describes the tissues, or the anatomical parts, and their role in function; this chapter describes the normal mechanics of the whole lumbar spine. Because most biomechanics texts provide descriptions of spine motion, I address that only briefly here. Instead, I focus on the functional implications of that motion, which are far more important. I also explain injury mechanisms and the changes that follow injury. Controversy remains as to whether these changes are a consequence of injury or in fact play a causative role. In addition, muscle activation challenges, spine loads, and in some cases stability are assessed over a variety of rehabilitation exercises and activities. This enables better decisions for therapeutic program design, because you will know what your clients should avoid and what they should perform.

Upon completion of this chapter, you will be able to explain the role of tissues in various tasks and consequently identify back-sparing techniques. In addition, you will understand the changes that follow injury, which have an impact on functional ability and rehabilitation decisions.

Kinematic Properties of the Thoracolumbar Spine

The ranges of thoracic and lumbar segmental motion about the three principal axes (shown in [table 4.1](#)) demonstrate the greater flexion, extension, and lateral bending capability of the lumbar region and the relatively greater twisting capability of the thoracic region. The segmental ranges shown in the table are population averages, but keep in mind that a large variability exists among people, among age groups (McGill, Yingling, and Peach, 1999), and among segments within an individual.

Table 4.1 Range of Motion of Each Spine Level (in Degrees)

Level	Flexion	Flexion and extension combined	Extension	Lateral bending	Axial twist
T1-2		4		6	9
T2-3		4		6	8
T3-4		4		6	8
T4-5		4		6	8
T5-6		4		6	8
T6-7		5		6	8
T7-8		6		6	8
T8-9		6		6	7
T9-10		6		6	4
T10-11		9		7	2
T11-12		12		9	2
T12-L1		12		8	2
L1-2	8		5	6	2
L2-3	10		3	6	2
L3-4	12		1	8	2
L4-5	13		2	6	2
L5-S1	9		5	3	5

All data are from White and Panjabi (1978), except flexion and extension lumbar data, which are from Pearcy, Portek, and Shepherd (1984) and Pearcy and Tibrewal (1984).

Joint stiffness values convey the amount of translational and rotational deformation of a spine section under the application of force or moment. The average stiffness values (shown in [table 4.2](#)) document the translational stiffness of the spine in a neutral posture; they indicate greater stiffness under compression than under shear loads. In rotational modes, greater stiffness occurs during axial torsion than during rotation about the flexion–extension and lateral bending axes. Although generally, the range of motion decreases with age, certain injuries, particularly to the disc, can increase the range of motion in bending and shear translation (Spencer, Miller, and Schultz, 1985). Kirkaldy-Willis and Burton (1992) implicated these large unstable movements in facet joint derangement. Recent data have quantified the increase in the range of motion about all three spine axes as disc degeneration proceeds from grade I to grades III and IV. Radial tears of the annulus are most prevalent in these stages. But this extra motion is replaced by extreme loss of motion in grade V discs, which are characterized by collapse and osteophyte formation (Tanaka et al., 2001).

Table 4.2 Average Stiffness Values for the Adult Human Spine

Spine level	Comp.	>SHEAR		BENDING		Axial torsion
		Ant./Post.	Lat.	Flex./Ext.	Lat.	
T1-T12	1,250	86/87	101	155/189	172	149
L1-L5	667	145/143	132	80/166	92	395
L5-S1	1,000	78/72	97	120/172	206	264

Compression and shear values are given in newtons per millimeter and bending and axial torsion in newton-meters per radian.

T1-T12 data are from White and Panjabi (1978). L1-L5 data are from Schultz et al. (1979) and Berkson, Nachemson, and Schultz (1979). L5-S1 data from McGlashen et al. (1987).

Motion Palpation—Pathology or Normal Asymmetry?

Clinical Relevance

When assessing a patient, stiffness asymmetries during bending to the right compared to the left and during twisting clockwise compared to counterclockwise at specific vertebral levels are not uncommon. This finding is of great importance to the clinician, who may suspect pathology at a specific location when what is present is simply normal anatomical asymmetry.

Our work (Ross, Bereznik, and McGill, 1999) exemplifies the peril in assuming that a joint with an asymmetrical feel upon palpation is pathological. Clinicians who hold to a typical motion palpation philosophy often identify an abnormal feeling at a specific spinal level as the target for therapy. Sometimes the asymmetrical stiffness is simply asymmetrical skeletal anatomy—perhaps a single facet with a unique angular orientation. Obviously, such a joint would be resistive to any mobilizing therapy.

As the spine moves in three dimensions (flexion–extension, lateral bend, and twist), the alignment of muscle vectors changes with respect to the vertebral orthopedic axes. This causes the role of the muscles to change. Sometimes their relative contribution to producing a specific moment changes along with the resultant joint compression and shear. Muscle lengths and their moment potentials as a function of spine posture are shown in [table 3.2](#) in chapter 3 and [table 4.3](#) here. A range of extreme postures is shown in [figure 4.1](#). Some muscles close to the spine obviously do not undergo great length excursions.

Table 4.3 Moment Potential (N · m) of Some Representative Muscles in Postures Shown in Figure 4.1

Muscle	FORCE (N)	UPRIGHT STANDING			60° FLEXION			25° RIGHT LATERAL BENDING			10° TWIST			COMBINED*		
		F ^b	B ^c	T ^d	F	B	T	F	B	T	F	B	T	F	B	T
R rectus abdominis	350	-28	17	6	-35	17	12	-28	21	8	-29	15	4	-33	20	15
R external oblique	315	-7	27	9	-7	24	15	-7	28	11	-8	28	6	-1	20	22
R internal oblique (L1)	280	-15	19	-16	-9	8	-31	-13	18	-20	-14	17	-18	-2	2	-33
R pars lumborum (L1)	455	36	23	9	35	23	8	36	24	9	3	24	5	38	23	7
R pars lumborum (L4)	595	23	23	-7	23	24	-4	23	23	-8	23	22	-8	23	25	-6
R iliocostalis lumborum	210	18	15	-1	18	14	2	18	15	-1	18	15	-2	18	15	1
R longissimus thoracis	280	24	10	-3	22	10	0	24	13	-3	24	12	-5	24	11	-1
R quadratus lumborum	175	5	11	5	7	14	1	5	12	6	6	13	4	7	14	0
R latissimus dorsi (L5)	140	9	5	-3	8	7	-4	8	6	-5	9	5	-5	8	7	-6
R multifidus	98	5	1	3	5	1	2	5	1	3	5	1	2	5	1	2
R psoas (L1)	154	1	9	3	-5	9	2	1	9	4	1	6	3	-4	8	4

Only a portion of some muscles are represented; for example, the laminae of psoas to L1 is shown rather than the whole psoas.

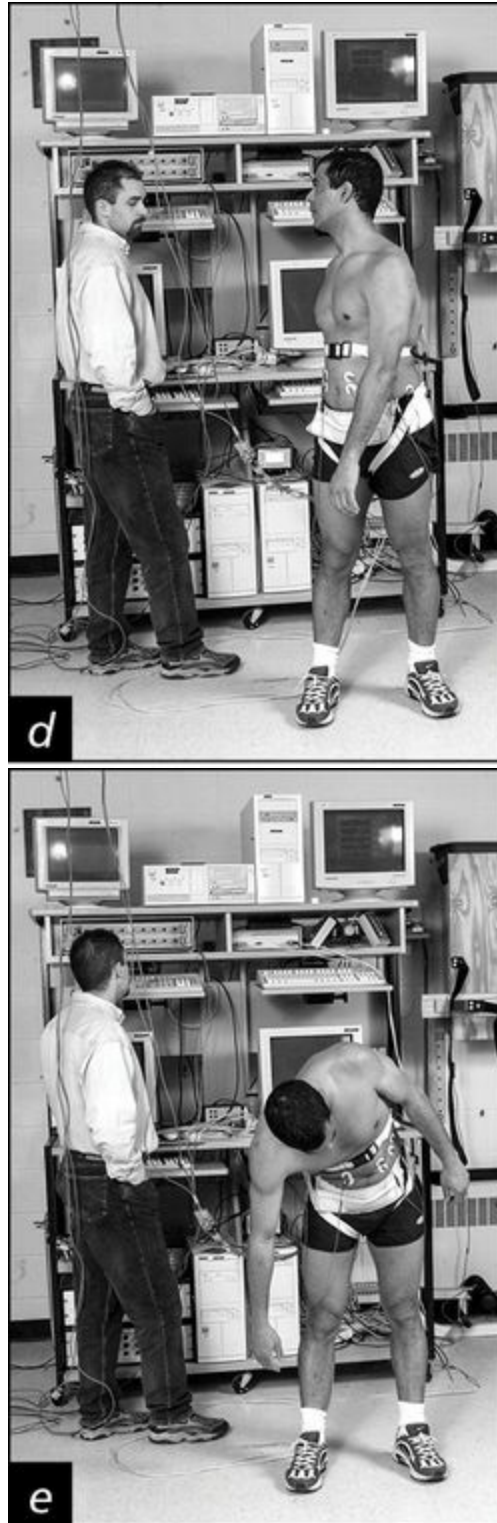
*Combination of 60° flexion, 25° right lateral bending, and 10° counterclockwise twist in one posture; ^bF = flexion; ^cB = lateral bend; ^dT = axial twist.

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Figure 4.1 A range of extreme postures was chosen to assess muscle length changes (table 4.3) and their potential to produce three-dimensional moments (table 4.3). Postures depicted are (a) upright standing, (b) 60° flexion, (c) 25° right lateral bending, (d) 10° twist, and (e) combined.





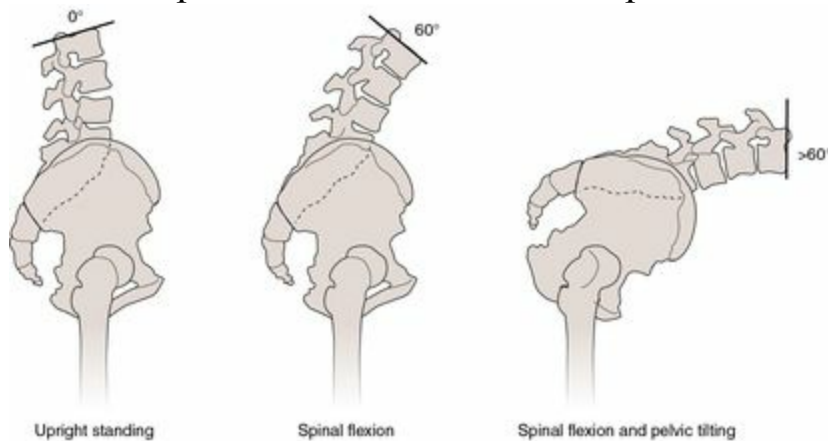


Photos from Stuart McGill

Lumbopelvic Rhythm: Questionable Assumptions

The typical description of torso flexion suggests that the first 60° takes place in the lumbar spine, whereas any further flexion is accomplished by flexion about the hips (see [figure 4.2](#)). Although this notion is very popular in clinical textbooks, we have never measured this strict sequence in anyone. In fact, Olympic weightlifters attempt to do the opposite—they lock the lumbar spine close to the neutral position and rotate almost entirely about the hips. When the lumbar spine and hip interplay is quantified in most people, it is apparent that torso flexion is accomplished with a combination of hip motion and lumbar spine flexion. In fact, given the ligament and annulus stresses associated with lumbar flexion, avoidance of full spine flexion is both prophylactic and therapeutic for most patients. The belief that the lumbopelvic rhythm (with distinct separation of spine and pelvic motion) is beneficial, as described in so many textbooks, appears to be a clinical myth and not the product of quantified spine and pelvis motion.

Figure 4.2 The lumbopelvic rhythm is the textbook description of how people bend over. It has been assumed that the first 60° takes place in the lumbar spine (flexing), and that further rotation of the torso is accomplished with rotation about the hip. We have never measured this sequence in anyone—from professional athlete to back patient.



Kinetics and Normal Lumbar Spine Mechanics

An interpretation of the function of the anatomical components of the lumbar spine requires the following:

- An analysis of their architecture and neural activation of muscles
- Knowledge of forces in the individual tissues (both active and passive) during a wide variety of tasks

This information is crucial for understanding how tissue overloading and injuries occur and also for optimizing treatment strategies for specific spine injury. [Table 4.4](#) and [figure 4.3](#) provide activation levels quantified with surface and intramuscular electromyography (EMG) for a variety of torso muscles and over a variety of activities. These are referred to throughout this book. This section addresses several issues and controversies about the functional interpretation of the thoracolumbar anatomy. Given the inability of the clinician and scientist to measure individual tissue forces in vivo, the only tenable option is to use sophisticated measurement techniques to collect biological signals from living subjects and integrate them with sophisticated modeling approaches to estimate tissue loads. Chapter 1 includes a brief description of the technique we used to assess the issues in this chapter (the virtual spine).

Table 4.4 Subject Averages of EMG Activation Normalized to Activity Observed During a Maximal Effort (100%)—Mean and (Standard Deviation in Parentheses)

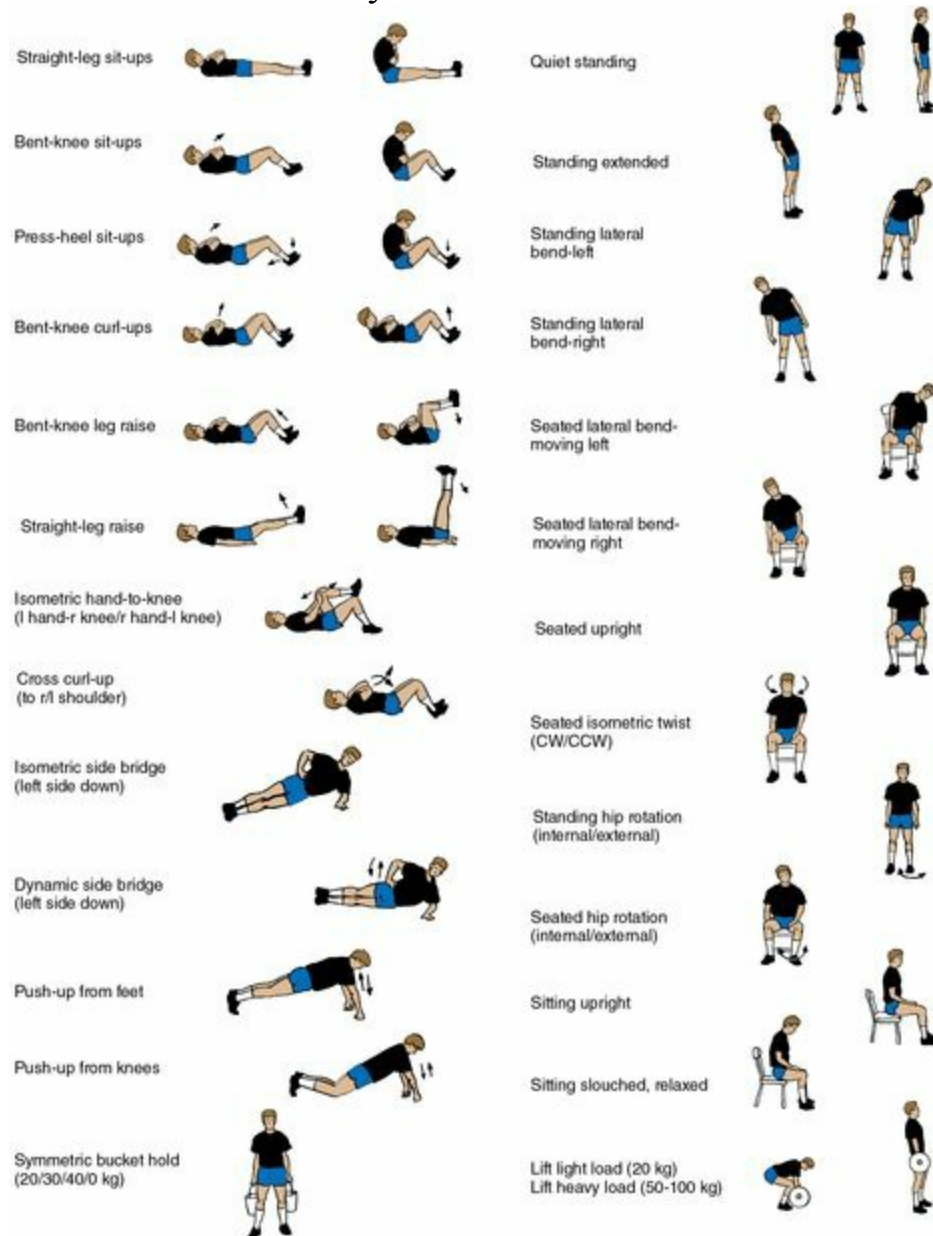
Task	PsoasTab1	PsoasTab2	EOI	IOI	TAI	RAs	RFs	ESs
Straight-leg sit-up	15 (±12)	24 (±7)	44 (±9)	15 (±15)	11 (±9)	48 (±18)	16 (±10)	4 (±3)
Bent-knee sit-up	17 (±10)	28 (±7)	43 (±12)	16 (±14)	10 (±7)	55 (±16)	14 (±7)	6 (±9)
Press-heel sit-up	28 (±23)	34 (±18)	51 (±14)	22 (±14)	20 (±13)	51 (±20)	15 (±12)	4 (±3)
Bent-knee curl-up	7 (±8)	10 (±14)	19 (±14)	14 (±10)	12 (±9)	62 (±22)	8 (±12)	6 (±10)
Bent-knee leg raise	24 (±15)	25 (±8)	22 (±7)	8 (±9)	7 (±6)	32 (±20)	8 (±5)	6 (±8)
Straight-leg raise	35 (±20)	33 (±8)	26 (±9)	9 (±8)	6 (±4)	37 (±24)	23 (±12)	7 (±11)
Isom. hand-to-knee (left hand → right knee, right hand → left knee)	16 (±16) 56 (±28)	16 (±8) 58 (±16)	68 (±14) 53 (±12)	30 (±28) 48 (±23)	28 (±19) 44 (±18)	69 (±18) 74 (±25)	8 (±7) 42 (±29)	6 (±4) 5 (±4)
Cross curl-up (right shoulder → across, left shoulder → across)	5 (±3) 5 (±3)	4 (±4) 5 (±5)	23 (±20) 24 (±17)	24 (±14) 21 (±16)	20 (±11) 15 (±13)	57 (±22) 58 (±24)	10 (±19) 12 (±24)	5 (±8) 5 (±8)
Isom. side support (left side down)	21 (±17)	12 (±8)	43 (±13)	36 (±29)	39 (±24)	22 (±13)	11 (±11)	24 (±15)
Dyn. side support (left side down)	26 (±18)	13 (±5)	44 (±16)	42 (±24)	44 (±33)	41 (±20)	9 (±7)	29 (±17)
Push-up from feet	24 (±19)	12 (±5)	29 (±12)	10 (±14)	9 (±9)	29 (±10)	10 (±7)	3 (±4)
Push-up from knees	14 (±11)	10 (±7)	19 (±7)	7 (±9)	8 (±8)	19 (±11)	5 (±3)	3 (±4)
Lift light load (20 kg, or 44 lb)	9 (±10)	3 (±4)	3 (±3)	6 (±7)	6 (±5)	14 (±21)	6 (±5)	37 (±13)
Lift heavy load (50-110 kg, or 110-243 lb)	16 (±18)	5 (±6)	5 (±4)	10 (±11)	10 (±9)	17 (±23)	6 (±5)	62 (±12)
Symmetrical bucket hold, 20 kg (44 lb)	2 (±4)	1 (±1)	7 (±4)	5 (±3)	5 (±1)	10 (±7)	3 (±3)	3 (±6)
40 kg (88 lb)	3 (±4)	1 (±1)	9 (±5)	6 (±4)	6 (±1)	10 (±8)	3 (±3)	4 (±7)
0 kg	3 (±5) 1 (±2)	1 (±1) 0 (±1)	10 (±6) 2 (±1)	8 (±6) 2 (±2)	6 (±2) 2 (±1)	10 (±8) 10 (±9)	3 (±3) 2 (±1)	3 (±2) 2 (±1)
Seated isom. twist CCW	30 (±20)	17 (±15)	18 (±8)	43 (±25)	49 (±35)	17 (±22)	7 (±4)	14 (±6)
Seated isom. twist CW	23 (±20)	11 (±8)	52 (±13)	15 (±11)	18 (±19)	13 (±10)	9 (±10)	13 (±8)
Standing hip internal rotation	21 (±18)	10 (±9)	18 (±12)	24 (±23)	33 (±20)	13 (±9)	9 (±7)	18 (±6)
Standing hip external rotation	27 (±20)	22 (±19)	17 (±13)	21 (±19)	31 (±17)	13 (±8)	19 (±11)	17 (±9)
Seated hip internal rotation	19 (±15)	21 (±18)	36 (±31)	30 (±30)	31 (±29)	18 (±8)	20 (±19)	12 (±8)
Seated hip external rotation	32 (±25)	25 (±20)	11 (±9)	15 (±17)	16 (±13)	15 (±9)	16 (±13)	8 (±8)
Seated upright	12 (±7)	7 (±5)	3 (±6)	3 (±3)	4 (±2)	17 (±9)	4 (±2)	5 (±8)
Seated slouched/relaxed	4 (±4)	3 (±3)	2 (±5)	2 (±2)	4 (±3)	17 (±11)	3 (±2)	5 (±8)
Quiet standing	2 (±1)	1 (±1)	3 (±4)	5 (±3)	4 (±2)	5 (±5)	3 (±3)	11 (±11)
Standing extended	3 (±2)	2 (±1)	12 (±9)	6 (±3)	5 (±3)	11 (±5)	4 (±3)	7 (±8)
Standing lateral bend, left	9 (±10)	1 (±2)	11 (±8)	18 (±14)	12 (±7)	13 (±7)	3 (±2)	11 (±13)
Standing lateral bend, right	6 (±5)	1 (±2)	19 (±18)	18 (±14)	25 (±20)	14 (±9)	3 (±1)	8 (±8)
Seated lateral bend, moving left	2 (±3)	1 (±1)	21 (±19)	7 (±7)	7 (±11)	13 (±8)	4 (±3)	6 (±7)
Seated lateral bend, moving right	18 (±12)	12 (±2)	15 (±26)	10 (±7)	12 (±7)	17 (±20)	5 (±4)	5 (±8)
Upright	14 (±9)	8 (±4)	6 (±4)	5 (±3)	5 (±5)	19 (±23)	5 (±3)	6 (±8)

Psoas channels, external oblique, internal oblique, and transverse abdominis are intramuscular electrodes, whereas rectus abdominis, rectus femoris, and erector spinae are surface electrodes.

EOI: external oblique (intramuscular); IOI: internal oblique (intramuscular); TAI: transverse abdominis (intramuscular); RAs: rectus abdominis; RFs: rectus femoris; ESs: erector spinae; ISOM: isometric; DYN: dynamic; CW: clockwise; CCW: counterclockwise

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Figure 4.3 Schematic documenting tasks during which EMG signals were obtained. They are listed in table 4.4.



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Loads on the Low Back During Functional Movements

Of course, no one can avoid performing countless functional movements

every day. And if your patient's form is poor for any of those movements, he is exacerbating his low back problems simply by going about his business. Thus, you must be able to analyze how your patient moves in all sorts of ordinary situations so that you can identify and explain where his problems lie, and how to correct them.

Standing and Then Bending Forward

People with discogenic disorders often become symptomatic with prolonged static postures, standing being one of them. Is early onset of pain during standing a warning sign of impending symptoms? Nelson-Wong and Callaghan (2014) found that those without clinical signs who developed transient pain when standing developed clinical low back pain in the future. Several hypotheses could account for this.

Several studies over the years have shown the flexion–relaxation phenomenon, or the apparent myoelectric silence of the low back extensor muscles during a standing-to-full-flexion maneuver. The hypothesis has been that, as full flexion occurs, either the extensors shut down their neural drive by reflex or the passive tissues simply take the load as they strain under full flexion. A study by McGill and Kippers (1994) using the virtual spine approach described in chapter 1 quantified individual tissue forces, thus adding more insight into this task. As one bends forward, the spine flexes and the extensors undergo eccentric contraction. As full flexion is approached, the passive tissues rapidly take over moment production, relieving the muscles of this role and accounting for their myoelectric silence. [Figure 4.4](#) shows the relative contribution of the muscles and the passive tissues (ligaments, disc, and gut) to the reaction moment throughout the movement. [Table 4.5](#) documents the distribution of tissue forces and their moments and joint load consequence. Interestingly, the relaxation of the lumbar extensor muscles appeared to occur only in an electrical sense because they generated substantial force elastically during full spine flexion through stretching. Perhaps the term flexion–relaxation is inappropriate, particularly for those who may be attempting to minimize forces in the muscle in clinical settings. Furthermore, the shear loading is substantial (see chapter 3 for a discussion of the ligament and muscle directions and the loss of shear support in the extensors with full flexion) and would suggest caution for those with spondylolisthesis or other subtler shear instabilities. Clearly, straight-leg toe

touches or knees-to-chest stretches would cause similar concern.

Figure 4.4 During standing to full forward flexion, then back to standing, the extensor muscles eccentrically contract but transfer their moment supporting role (MuscleTq) to the passive tissues at full flexion—the disc (DiscTq), the buckled gut (GutTq), and the ligaments (LigTq). Note that some force remains in the muscles with passive stretching.

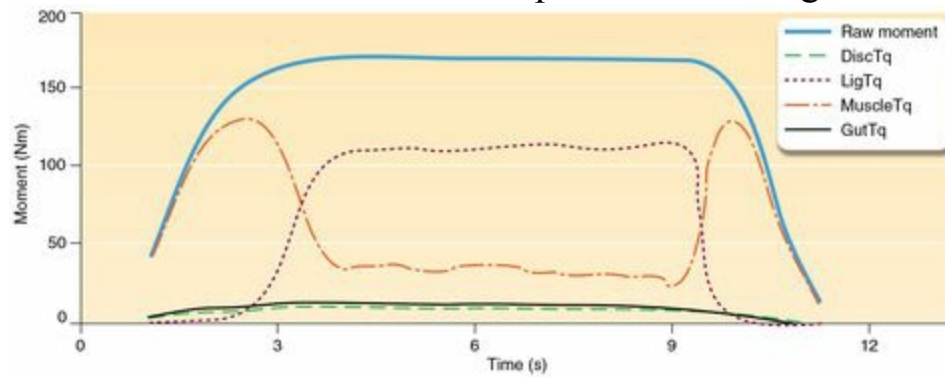


Table 4.5 Individual Muscle and Passive Tissue Forces and Moments During Full Flexion

	FORCE	MOMENT (NM)			COMPRESSION	SHEAR (N)	
	N	Flexion	Lateral	Twist	N	Anteroposterior	Lateral
MUSCLE							
R rectus abdominis	16	-2	1	1	15	5	-4
L rectus abdominis	16	-2	-1	-1	15	5	4
R external oblique 1	10	-1	1	1	8	7	-3
L external oblique 1	10	-1	-1	-1	8	7	3
R external oblique 2	7	-1	1	0	6	2	-3
L external oblique 2	7	-1	-1	0	6	2	3
R internal oblique 1	35	0	3	-2	21	-19	20
L internal oblique 1	35	0	-3	2	21	-19	-20
R internal oblique 2	29	-2	2	-3	8	-17	21
L internal oblique 2	29	-2	-2	3	8	-17	-21
R pars lumborum (L1)	21	2	1	0	21	6	2
L pars lumborum (L1)	21	2	-1	0	21	6	-2
R pars lumborum (L2)	27	2	1	0	26	8	2
L pars lumborum (L2)	27	2	-1	0	26	8	-2
R pars lumborum (L3)	31	1	1	0	29	-4	6
L pars lumborum (L3)	31	1	-1	0	29	-4	-6
R pars lumborum (L4)	32	1	1	0	30	-7	6
L pars lumborum (L4)	32	1	-1	0	30	-7	-6
R iliocostalis lumborum	58	5	4	1	57	14	-1
L iliocostalis lumborum	58	5	-4	-1	57	14	1
R longissimus thoracis	93	7	4	0	91	23	-6
L longissimus thoracis	93	7	-4	0	91	23	6
R quadratus lumborum	25	1	2	0	25	-1	1
L quadratus lumborum	21	1	-2	0	25	-1	-1
R latissimus dorsi (L5)	15	1	1	0	14	-1	-6
L latissimus dorsi (L5)	15	-1	1	0	14	-1	6
R multifidus 1	28	1	1	1	26	6	9
L multifidus 1	28	1	-1	-1	26	6	-9
R multifidus 2	28	1	1	0	28	6	0
L multifidus 2	28	1	-1	0	28	6	0
R psoas (L1)	25	1	2	0	24	9	6
L psoas (L1)	25	-1	-2	0	24	9	-6
R psoas (L2)	25	-1	2	0	24	9	6
L psoas (L2)	25	0	-2	0	24	9	-6
R psoas (L3)	25	0	1	0	24	9	7
L psoas (L3)	25	0	-1	0	24	9	-7
R psoas (L4)	25	0	1	1	24	9	8
L psoas (L4)	25	0	-1	-1	24	9	-8

	FORCE	MOMENT (NM)			COMPRESSION	SHEAR (N)	
	N	Flexion	Lateral	Twist	N	Anteroposterior	Lateral
MUSCLE							
Anterior longitudinal	0	0	0	0	0	0	—
Posterior longitudinal	86	2	0	0	261	44	—
Ligamentum flavum	21	1	0	0	21	2	—
R intertransverse	14	0	0	0	13	3	—
L intertransverse	14	0	0	0	13	3	—
R articular	74	2	1	1	65	40	—
L articular	74	2	-1	-1	65	40	—
R articular 2	103	3	2	2	84	-3	—
L articular 2	103	3	-2	-2	84	-3	—
Interspinous 1	301	18	0	0	273	142	—
Interspinous 2	345	14	0	0	233	268	—
Interspinous 3	298	10	0	0	194	238	—
Supraspinous	592	41	0	0	591	79	—
R lumbodorsal fascia	122	8	1	0	109	-1	—
L lumbodorsal fascia	122	8	-1	0	109	-1	—
PASSIVE							
Disc	—	9	0	0	—	—	—
Gut, etc.	—	11	0	0	—	—	—

The moment was 171 Nm (38 Nm by muscle, 113 Nm by ligaments, and 20 Nm by passive tissues, such as the disc, skin, and buckled viscera). The joint compression was 3,145 N, and shear was 1,026 N.

Loads on the Low Back During Lifting

During lifting, the muscle and ligament forces required to support the posture and facilitate movement impose mammoth loads on the spine. This is why lifting technique is so important to reduce low back moment demands and the risk of excessive loading. The following example demonstrates this concept.

A man is lifting 27 kg (59.5 lb) held in the hands using a squat lift style. This produces an extensor reaction moment in the low back of 450 Nm. The forces in the tissues that support this moment impose a compressive load on the lumbar spine of over 7,000 N. [Table 4.6](#) details the contributions to the total extension moment and to the forces from the muscular components. These forces and their effects are predicted using the sophisticated modeling approach, which uses biological signals obtained directly from the subject (see chapter 1). It should be noted here that 7,000 N of compression begins to cause damage in very weak spines, although the tolerance of the lumbar spine in an average healthy young man probably approaches 12 to 15 kN (Adams and Dolan, 1995). In extreme cases, compressive loads on the spines of competitive weightlifters have safely exceeded 20 kN (Cholewicki, McGill, and Norman, 1991).

Table 4.6 Musculature Components for Moment Generation of 450 Nm During Peak Loading for a Squat Lift of 27 kg (59.5 lb)

Muscle	Force (N)	Moment (Nm)	Compression (N)	Shear (N)
Rectus abdominis	25	-2	24	5
External oblique 1	45	1	39	24
External oblique 2	43	-2	30	31
Internal oblique 1	14	1	14	-2
Internal oblique 2	23	-1	17	-16
Longissimus thoracis pars lumborum L4	862	35	744	-436
Longissimus thoracis pars lumborum L3	1,514	93	1,422	-518
Longissimus thoracis pars lumborum L2	1,342	121	1,342	0
Longissimus thoracis pars lumborum L1	1,302	110	1,302	0
Iliocostalis lumborum pars thoracis	369	31	369	0
Longissimus thoracis pars thoracis	295	25	295	0
Quadratus lumborum	393	16	386	74
Latissimus dorsi L5	112	6	79	-2
Multifidus 1	136	8	134	18
Multifidus 2	226	8	189	124
Psoas L1	26	0	23	12
Psoas L2	28	0	27	8
Psoas L3	28	1	27	6
Psoas L4	28	1	27	5

Negative moments correspond to flexion, whereas negative shear corresponds to L4 shearing posteriorly on L5.

Understanding the individual muscle forces, their contributions to supporting the low back, and their components of compression and shear force that are imposed on the spine is very useful. In this particular example, the lifter avoided full spine flexion by flexing at the hip, minimizing ligament and other passive tissue tension and relegating the moment generation responsibility to the musculature. An example in which the spine is flexed is presented later in this chapter. As described in chapter 3, the pars thoracis extensors are very effective lumbar spine extensors, given their large moment arms. Also, because the lifter's upper body is flexed, large reaction shear forces on the spine are produced (the rib cage is trying to shear forward on the pelvis). These shear forces are supported to a very large degree by the pars lumborum extensor muscles. Furthermore, the abdominal muscles are activated but do not produce movement. Why are they active? These muscles are activated to stabilize the spinal column, although this mild abdominal activity imposes a compression penalty to the spine. A more robust explanation of stabilizing mechanics is presented in chapter 5.

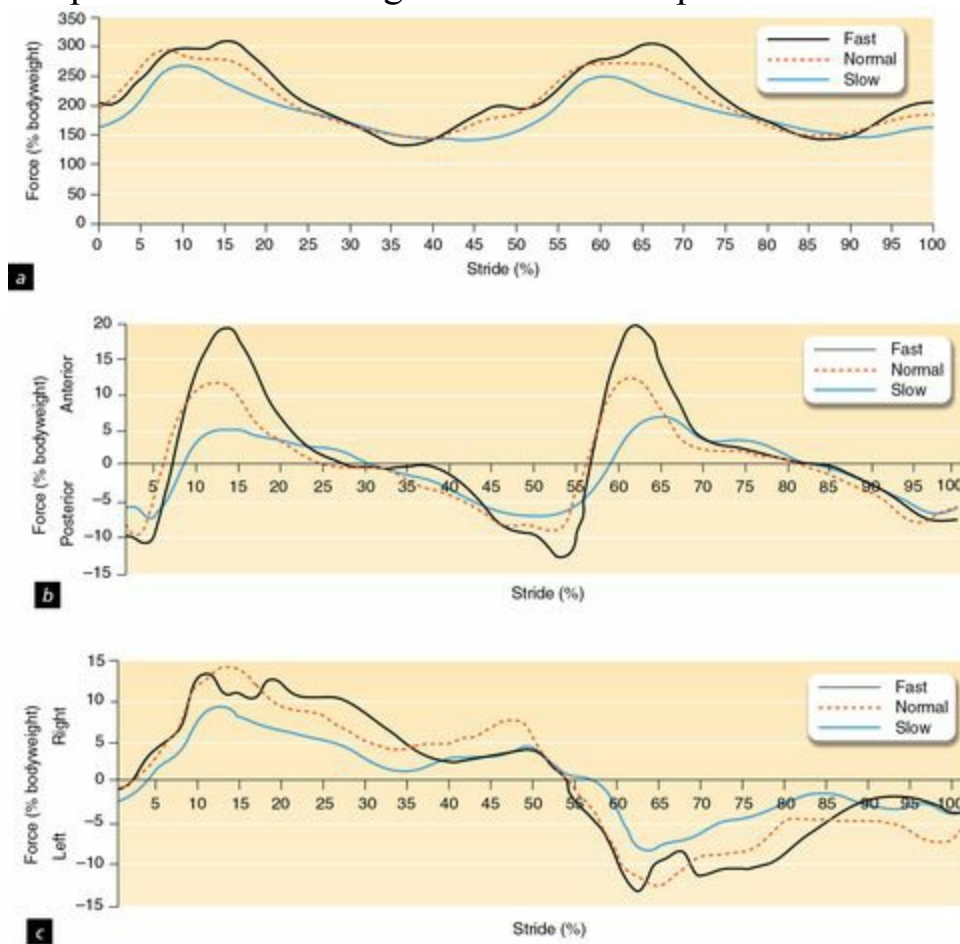
The preceding example demonstrates how diffusely the forces are distributed and illustrates how proper clinical interpretation requires anatomically detailed free body diagrams that represent reality (such as those

incorporated into the virtual spine model). I believe that oversimplified free body diagrams have resulted in clinicians' overlooking of the important mechanical compressive and especially shear components of muscular force. This has compromised the assessment of injury mechanisms and the formulation of optimal therapeutic exercise.

Loads on the Low Back During Walking

Thousands of low-level loading cycles are endured by the spine every day during walking. Although the small loads in the low back during walking suggest that the activity is safe and tolerable, clinicians have found that walking provides relief to some people but is painful to others. Recent work has suggested that walking speed affects spine mechanics and may account for these differences. During walking, the compressive loads on the lumbar spine of approximately 2.5 times body weight, together with the very modest shear forces, are well below any known in vitro failure load (see [figure 4.5](#)). Strolling reduces spine motion and produces almost static loading of tissues, however, whereas faster walking, with arms swinging, causes cyclic loading of tissues (Callaghan, Patla, and McGill, 1999) (see [figure 4.6](#)). This change in motion may begin to explain the relief experienced by some. Arm swinging while walking faster, with all other factors controlled, results in lower lumbar spine torques, muscle activity, and loading (see [figure 4.7](#)). In fact, we have observed up to 10% reduction in spine loads from arm swinging in some people. This may be because swinging the arms facilitates the efficient storage and recovery of elastic energy, reducing the need for concentric muscle contraction and the upper-body accelerations associated with each step. Interestingly, Kubo and colleagues (2006) reported higher torso stiffness with faster walking, which would further facilitate efficient energy recovery. Also interesting is the fact that fast walking has been shown to be a positive cofactor in the prevention of, and more successful recovery from, low back troubles (Nutter, 1988).

Figure 4.5 Loads on the lumbar spine (normalized to body weight) during three speeds of walking and with normal arm swing. The curves are normalized for one stride (beginning and ending with right heel contact). (a) L4-L5 compression, (b) anteroposterior shear forces in which positive indicates anterior shear of the superior vertebrae, (c) lateral shear force in which positive indicates right shear of the superior vertebrae.



Reprinted from Clinical Biomechanics, 14, J.P. Callaghan, A.E. Patla, and S.M. McGill, "Low back three-dimensional joint forces, kinematics and kinetics during walking," 203-216, 1999, with permission from Elsevier Science.

Figure 4.6 Lumbar motion during three speeds of walking for one normal subject (beginning and ending with right heel contact): (a) lumbar flexion–extension in which extension is positive, (b) lateral bend in which positive indicates bending to the right, (c) axial twist in which positive indicates the upper body twisting to the right.

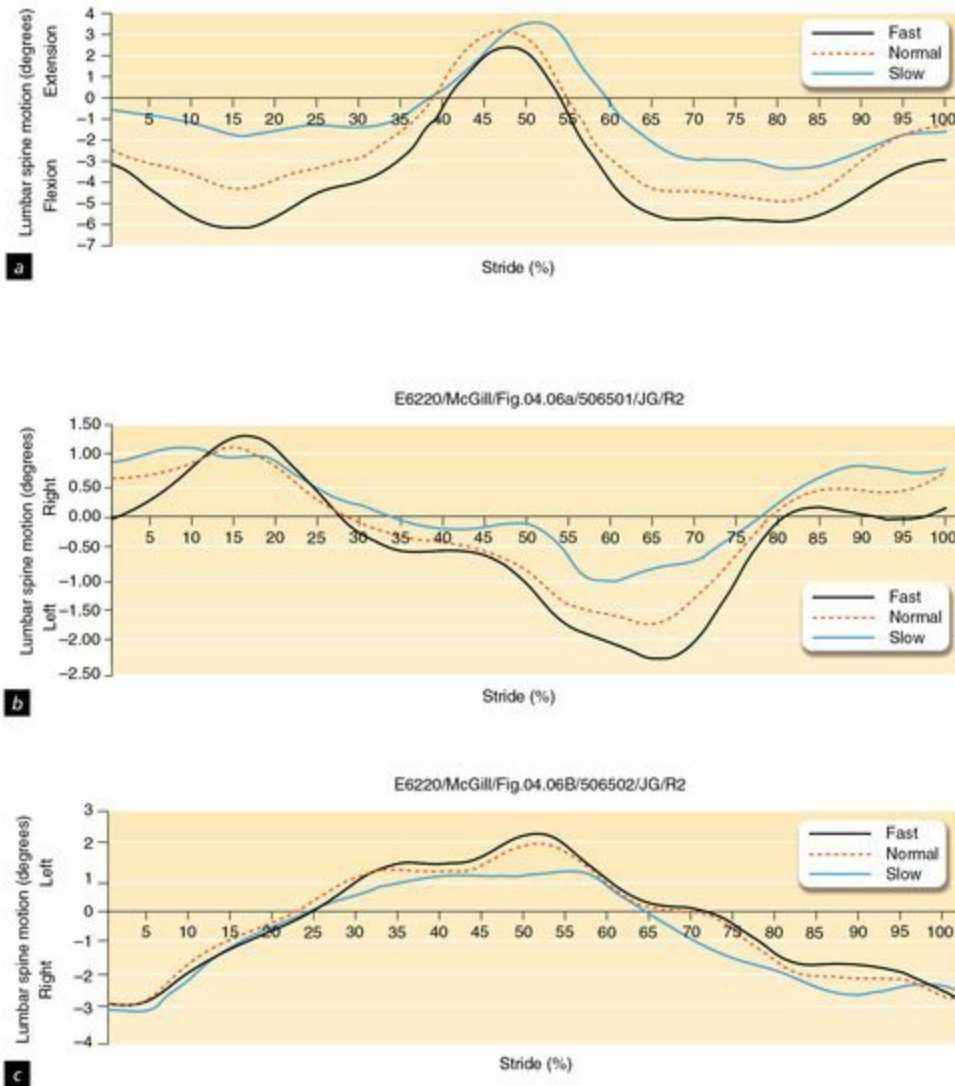
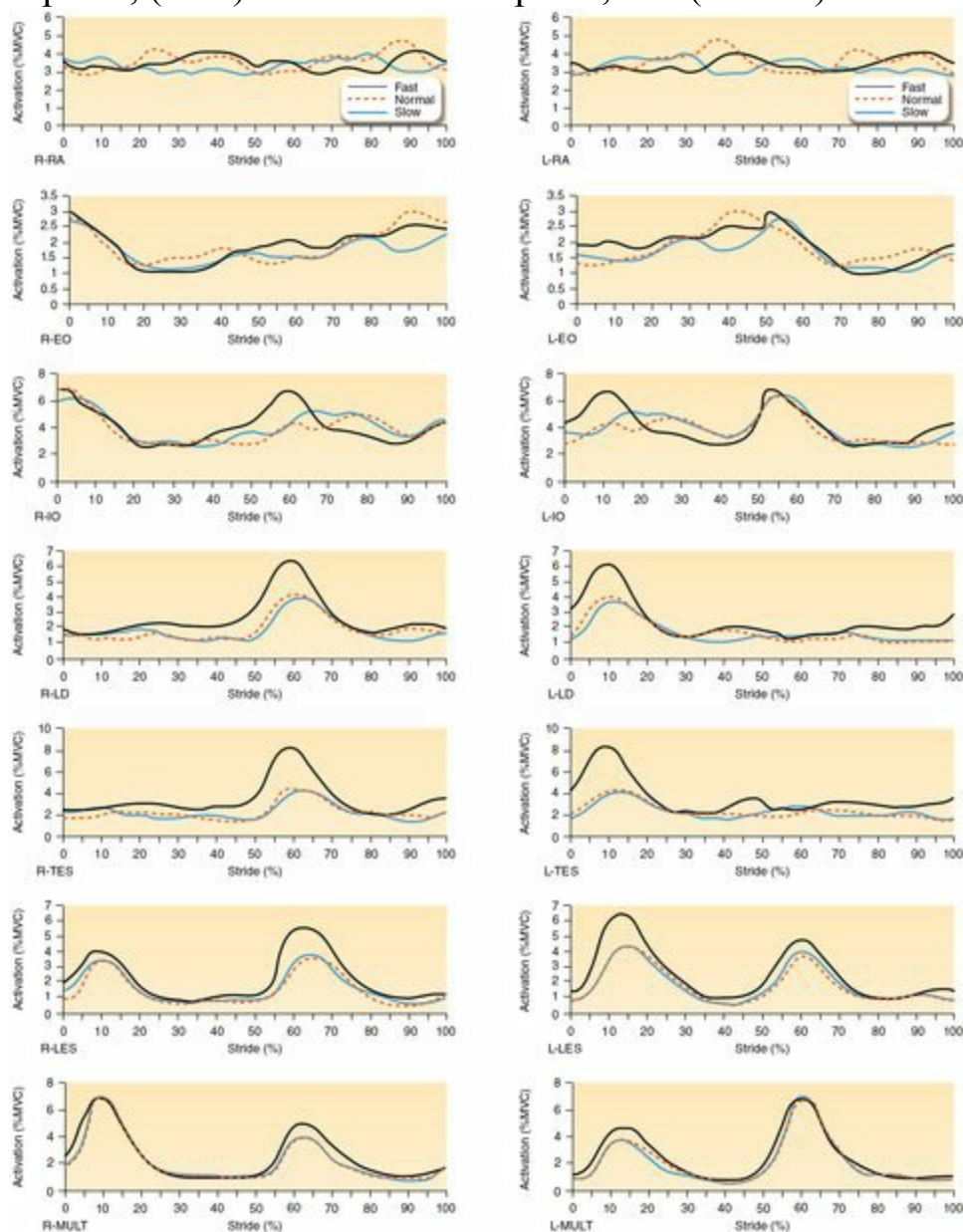


Figure 4.7 Activation profiles (EMG signals) from 14 torso muscles during three speeds of walking with normal arm swing (beginning and ending with right heel contact). The muscle pairs are (RA) rectus abdominis, (EO) external oblique, (IO) internal oblique, (LD) latissimus dorsi, (TES) thoracic erector spinae, (LES) lumbar erector spinae, and (MULT) multifidus.



Many patients are concerned about the implications of walking with shoes with elevated heels. Barton, Coyle, and Tinley (2009) documented the increased erector spinae muscle activity with heels elevated 20 cm (about 8 in.). Activity initially increased 20% immediately following heel strike, but

after 2 days of accommodating to the extra heel height, it increased to 24%. Explanations included a pitched-forward torso and changes in the shock absorption of a more flexed subtalar joint.

Therapeutic Walking

Clinical Relevance

Fast Walking

Fast walking is generally therapeutic (Nutter, 1988). Several mechanisms appear to account for this: reciprocal muscle activation and tissue load sharing, gentle motion, and reduced spine loads with energy conservancy from arm swing. In contrast, these benefits do not occur during slow walking or mall strolling, which exacerbates symptoms in many because of the static loading that results. Finally, we have noted that people with pained backs exemplify the typical pained general flexor response in that they often tend to swing the arms primarily about the elbows. This should be corrected to arm swing about the shoulders to optimize the benefits of arm swing.

Aquatherapy

Many assume that walking in a pool (aquatherapy) unloads the joints because weight is supported with water buoyancy. With no movement, this is correct. But this is an inappropriate assumption because the moving limbs require much greater joint moments to propel them against the viscous resistance of the water. Kaneda and colleagues (2009) documented elevated leg and torso muscle activity during deep-water running and in-pool walking at three speeds compared with land walking—for example, erector spine activity during land walking of 11% of maximal voluntary contraction (MVC) versus 17% in the pool, and gluteus maximus activity on land of 9% versus 12% in the pool. There are other considerations and merits to aquatherapy, however. Water exercise assists in venous return of blood to the heart. The hydrostatic pressure of the water equals blood pressure at just over 1 m (3.3 ft) in depth so that the blood is returned to the heart "for free" with body parts submerged at these depths.

Loads on the Low Back During Sitting

Nachemson's classic work (1966), using intradiscal pressure measurements, documented the higher loads on the discs in various sitting postures compared to the standing posture. Normal sitting causes flexion in the lumbar spine, and people, if left alone, generally sit in a variety of flexed postures (Callaghan and McGill, 2001b). Sitting generally involves lower abdominal wall activity (particularly the deep abdominal muscles) compared to standing, and unsupported sitting generally involves higher extensor activity (see Callaghan, Patla, and McGill, 1999, for walking and Callaghan and McGill, 2001b, for sitting). Sitting slouched minimizes muscle activity, whereas sitting more upright requires higher activation of the psoas and the extensors (Juker, McGill, Kropf, and Steffen, 1998). Full flexion increases disc annulus stresses; although this posture has produced disc herniations in the lab (e.g., Wilder, Pope, and Frymoyer, 1988), substantial simultaneous load was needed. In fact, Kelsey (1975) discovered a specific link between prolonged sitting and the incidence of herniation. But association is not causation. We have not been able to initiate a herniation with sitting postures. Rather, the cumulative herniating trauma is caused in places such as the gym and as a result of poor lifting form throughout the day. Once the delaminations caused by exposure to flexion and load in the annulus have been initiated, then sitting adds to the trauma. Thus, those who cannot tolerate sitting for a period of time have already begun the cumulative trauma cascade. Paradoxically, those who have never stressed their spines usually find sitting comfortable.

Transitioning from a slouched to an upright posture is usually accomplished with two approaches ([figure 4.8](#)). Contrary to common perception, even when sitting upright, the L5/S1 joint is still quite flexed (60% of its neutral-to-full-flexion range, on average) (Dunk et al., 2009). When correcting this posture, some choose to lift the rib cage, hinging more at the thoracolumbar region with thoracic erector spinae activity. Not surprisingly, this strategy often results in midback pain and responds to corrective movement strategies such as the Lewit exercise (shown in chapter 10). Others instead focus more on flexing the hips, which aligns the spine above it. Comparing the two strategies, Casthanhero, Duarte, and McGill (2014) found roughly equivalent spine loading—thoracic erectors in the rib lift strategy and the psoas and hip flexors in the pelvic tilt strategy. A combination of the two was preferred by most people and resulted in the least spine loading. Thus, changing lumbar postures causes a migration of the

loads from one tissue to another. Sitting upright relieves pain due to this load migration. Callaghan and McGill (2001b) suggested that no single, ideal sitting posture exists for any length of time; rather, a variable posture is recommended to minimize the risk of tissue overload. For example, Snijders, Hermens, and Kleinrensink (2006) suggested that a cross-legged sitting posture stabilizes the sacroiliac joints via passive tensioning of the iliolumbar ligament and piriformis muscle.

Figure 4.8 Sitting with (a) a flexion slouch is corrected by two common strategies. (b) Lifting the ribcage creates a hinge at the thoracolumar junction often creating pain or sensitivity at this level, and (c) flexing the hips to achieve a neutral posture. The least stress is achieved with a combination of the two.





Loads from Backpack Carriage

Backpacks come in various designs that affect low back loading. Generally, if rough terrain is anticipated, the load should be placed low in the pack to minimize the moment arm or distance to the low back. As the load is carried over rough ground, it accelerates and decelerates. The load placed closer to the low back reduces the torso forces needed to move the backpack load. On the other hand, if smooth ground is anticipated, carrying the load high in the pack, and over the fulcrum of the low back and hips, requires smaller torso muscle forces—and lower lumbar spine loads result.

Backpack Carrying

Clinical Relevance

Now for the curious situation in which backpacks can reduce spine loads

—and thus form one of our exercise-based therapy prescriptions. A person who is flexion intolerant, and also has posterior discogenic back pain exacerbated by prolonged sitting, generally has difficulty standing up. Upon standing, a forward torso angle (antalgic posture) remains. If this person can tolerate compression, we prescribe wearing a backpack with about 10 kg (22 lb) placed low in the backpack (about the level of the lumbar spine) and going for a walk over uneven ground. Wearing the backpack generates torso extensor moment, bringing the torso into an upright posture. This alleviates the spine extensors, which were previously contracted in the standing, but flexed, posture. Given their larger moment arm, this reduces the compressive load on the spine. In fact, recent observations by Rohlmann and colleagues (2014) confirmed that carrying 9 kg (20 lb) in a backpack did not increase the load on the lumbar spine of an older man, which was detected via an instrumented vertebral body replacement. The compression reduction from the muscles shutting down is larger than the extra compression from the additional load in the pack, resulting in a net reduction in total compression on the back. Walking over uneven ground provides gentle motion to the lumbar spine, which is therapeutic to the type of discogenic person we are describing here. Typically, the patient returns saying, “Thanks—that was amazing.” Thus, although some have blamed backpacks as a source of back troubles, they can actually be used therapeutically.

I recall a radio interview in which a spine expert was claiming that carrying backpacks over one shoulder was a serious problem for children. The interviewers phoned me for my on-air comment. I explained that, no doubt, some children will experience troubles, but there is also a training opportunity here. If the children were to switch shoulders frequently, this problematic task would become clever back training! The issue of the training load versus the dangerous load hinges on some subtle modulators. Perhaps the spine expert was right—that the children should carry their backpacks on both shoulders. But the technique of changing shoulders would have transformed a perceived danger into training for better health and performance.

Loads on the Low Back During Various Exercises

Because exercise is a crucial element of rehabilitation for low back problems, it is crucial that you understand the loads you are imposing on your patient's back when you prescribe an exercise. Otherwise, what was intended to be therapeutic may exacerbate her back troubles. Mastering the information in this section will increase your ability to tailor every exercise to each client's unique needs not only at the outset of treatment, but at every stage of progression toward better low back health.

Loads on the Low Back During Flexion Motion Exercises

Very few studies (only our own that we are aware of) have quantified the tissue loading on the low back tissues during various types of torso flexion exercises, although some have measured the EMG activity of selected muscles (e.g., Flint, 1965; Halpern and Bleck, 1979; Jette, Sidney, and Cicutti, 1984). This type of information alone can provide insight into relative muscle challenges, but it is restrictive for guiding exercise prescription decisions because the resultant spine load is unknown. The goal is to challenge muscle at appropriate levels but in a way that spares the spine. Too many exercises are prescribed that exceed the tolerance of back sufferers' compromised tissues. In fact, I believe that many commonly prescribed flexion exercises result in so much spine compression that these exercises, together with the flexion motion stresses, will ensure that the person remains a patient. For example, the traditional sit-up imposes approximately 3,300 N of compression on the spine (Axler and McGill, 1997). ([Figure 4.9](#) illustrates psoas and abdominal muscle activation levels in a variety of flexion tasks. [Figure 4.10](#) illustrates activation patterns with bent-knee sit-ups, and [figure 4.11](#) illustrates activation patterns with bent-knee curl-ups.) Note that muscle activation levels are expressed in normalized units (%MVC). This means that the activity is expressed as a percentage of what would be observed during a maximal voluntary contraction (MVC), thus quantifying activity in a physiological and functional context. Further, the spine is very flexed during the period of this load (McGill, 1998). The National Institute for Occupational Safety and Health (NIOSH) (1981) in the United States has set the action limit for low back compression at 3,300 N; repetitive loading above this level is linked with higher injury rates in workers, yet this is imposed on the spine with each repetition of the sit-up!

Figure 4.9 Activation of the psoas and the abdominal muscles in a variety of flexion tasks from a group of highly trained subjects (five men and three women).

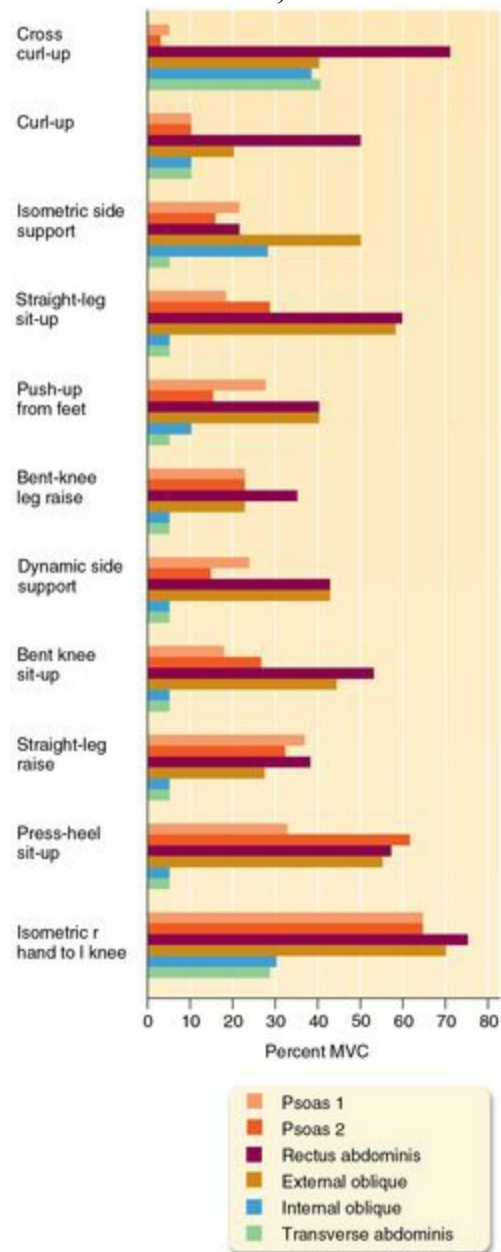


Figure 4.10 Activation-time histories of the same subjects as in figure 4.9 performing a bent-knee sit-up. Surface and indwelling electrodes are indicated.

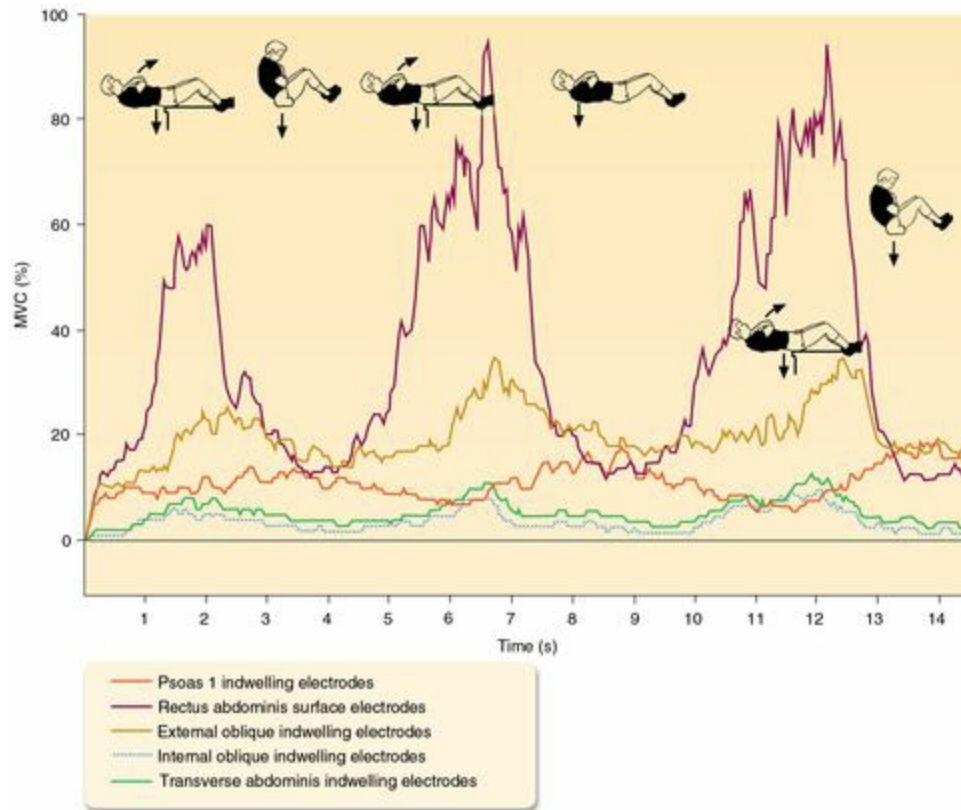
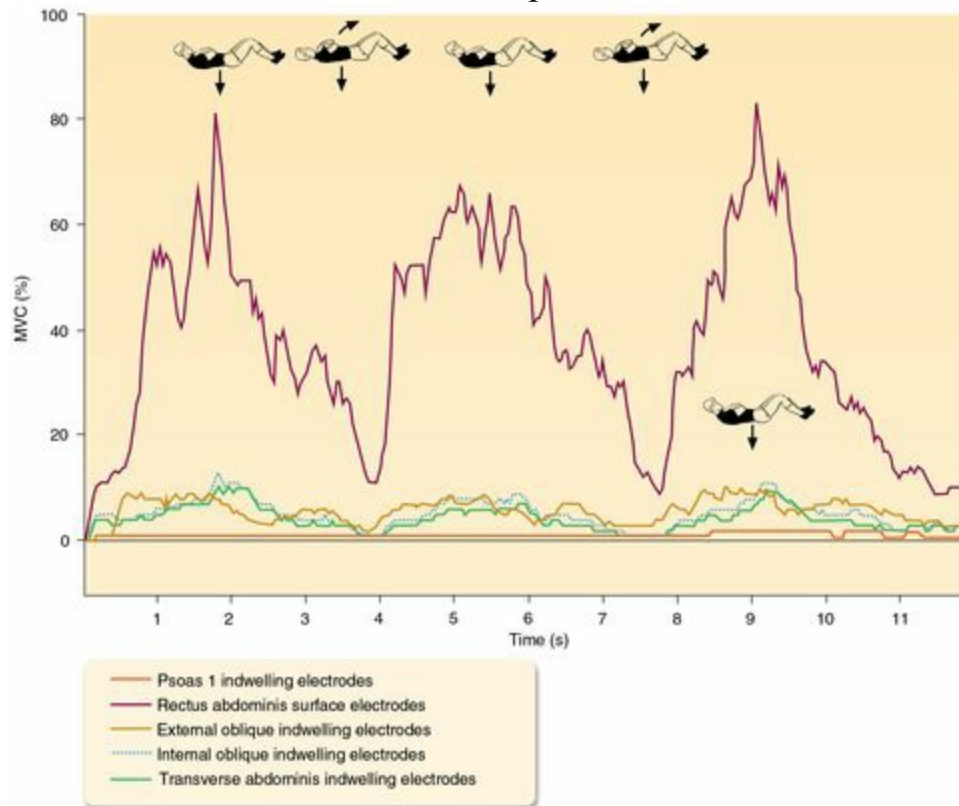


Figure 4.11 Activation-time histories for the same group performing bent-knee curl-ups.



Many recommend performing sit-ups with the knees bent, the theory being that the psoas is realigned to reduce compressive loading, or perhaps the psoas is shortened on the length–tension relationship so that the resulting forces are reduced. After examining both of these ideas, we found them to be untenable. We recruited a group of women who were small enough to fit into a magnetic resonance imaging (MRI) scanner. We placed each woman into the scanner and varied her knee and hip angles while she was supine (Santaguida and McGill, 1995). The psoas did not change its line of action, nor could it because it is attached to each vertebral body and transverse process (as the lumbar spine increases lordosis, the psoas follows this curve). The psoas does not change its role from a flexor to an extensor as a function of lordosis—this interpretation error occurred from models in which the psoas was represented as a straight-line puller. In fact, the psoas follows the lordotic curve as the lumbar spine flexes and extends. Further, it is true that the psoas is shortened with hip flexion, but its activation level is higher